

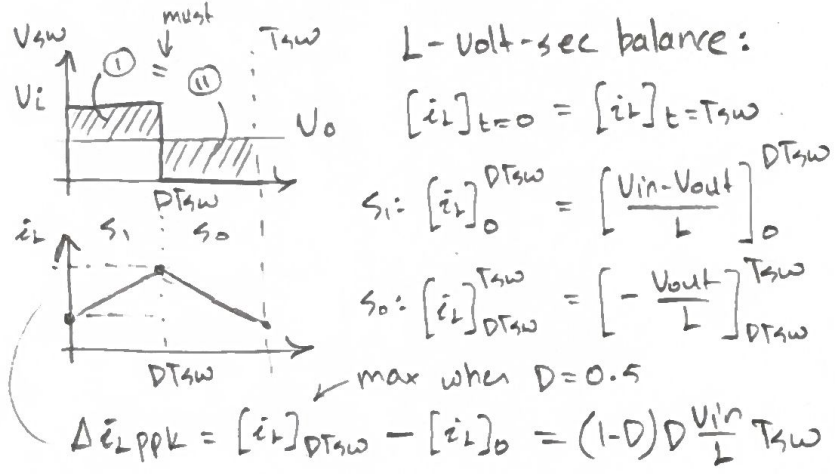
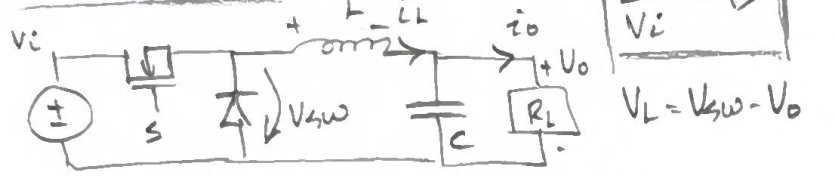
Power Efficiency: $\eta = \frac{P_{out}}{P_{in}} = \frac{P_{loss1}}{P_{loss2}} = \frac{P_{in} - P_{out1}}{P_{in} - P_{out2}} = \frac{1 - D_1}{1 - D_2}$

Linear Mode Converter:

$V_i \xrightarrow{R} V_o$ $\frac{V_o}{V_i} = \frac{R_{out}}{R + R_{out}}$, $\eta = \frac{R_{out}}{R + R_{out}}$

• efficiency is high only if V_{out} is close to V_{in}
 • poor regulation since V_{out} changes if R_{out} changes

Buck Converter



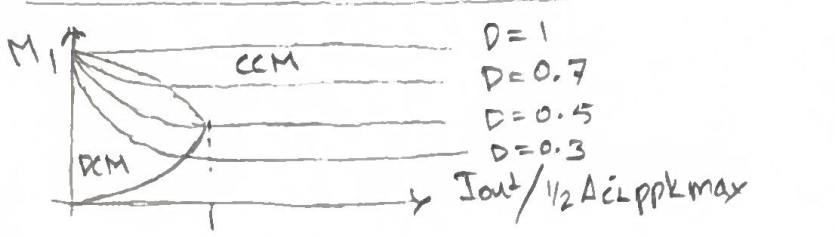
CCM: $\frac{1}{2} \Delta i_{Lppk} < I_{out}$

DCM: $\frac{1}{2} \Delta i_{Lppk} > I_{out}$

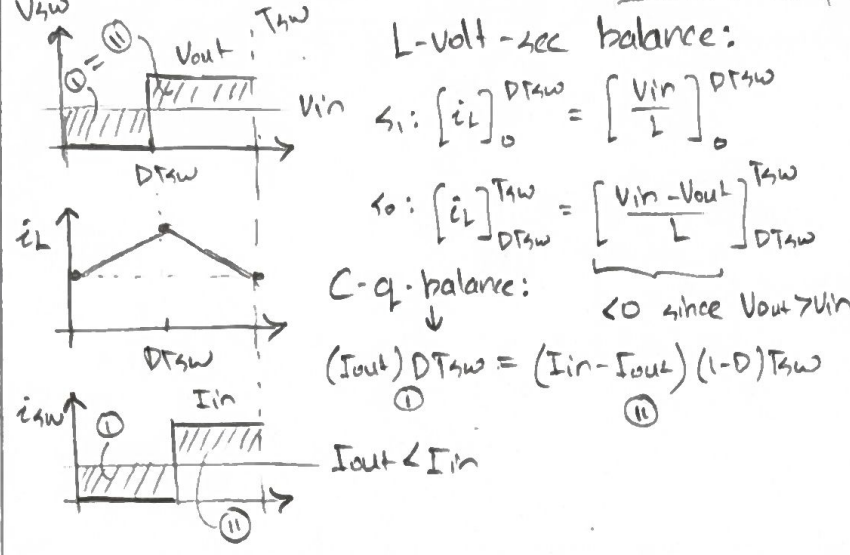
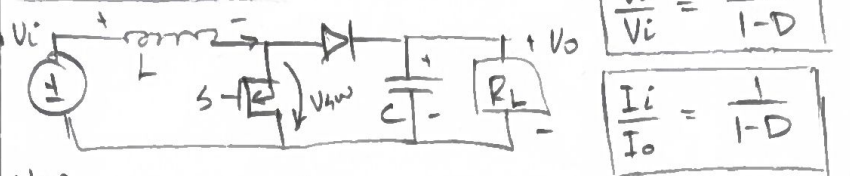
critical load resistance: $R_{out} > \frac{2L}{(1-D)T_{sw}}$

$\Delta i_{Lppk} = \left(\frac{V_{in} - V_{out}}{L} \right) DT_{sw}$

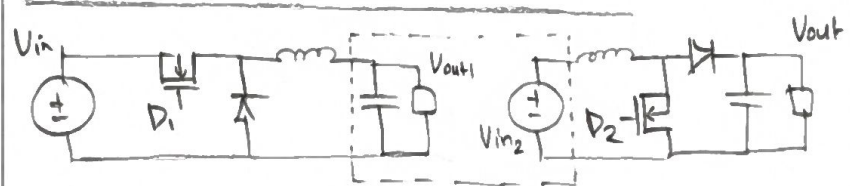
$M = \frac{2}{1 + \sqrt{1 + \frac{4K}{D^2}}}$, $K = \frac{2L}{T_{sw} \cdot R_{out}}$



Boost Converter



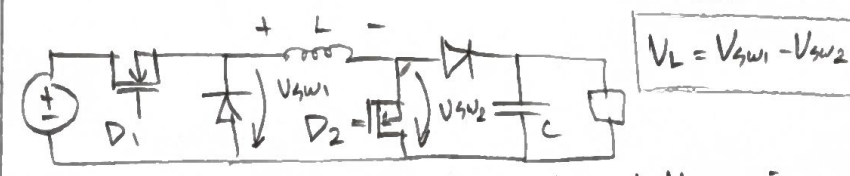
4-switch Buck Boost Converter



From Buck: $\frac{V_{out1}}{V_{in}} = D_1 \Rightarrow V_{out1} = D_1 V_{in}$

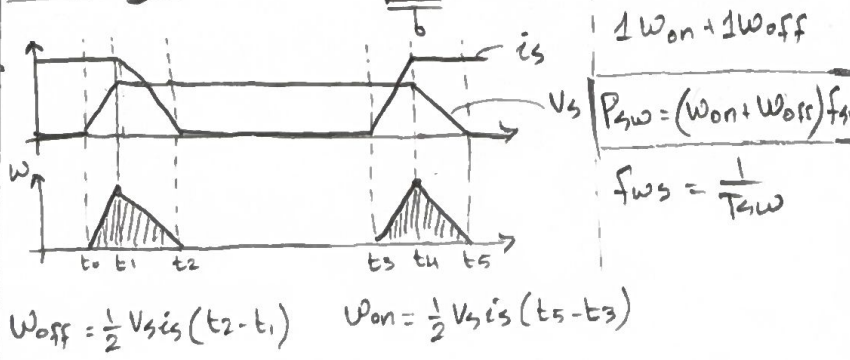
From cascade: $V_{in2} = V_{out1}$

From Boost: $\frac{V_{out}}{V_{in2}} = \frac{1}{1-D_2} \Rightarrow \frac{V_{out}}{V_{in}} = \frac{D_1}{1-D_2}$

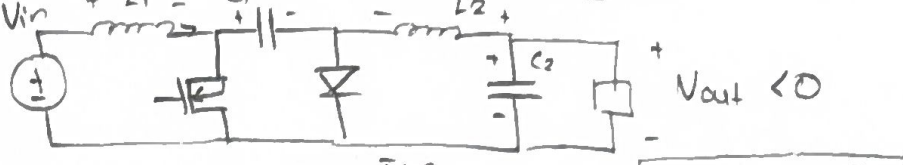


can vary V_{sw1} , V_{sw2} thru D_1 , D_2 to get $V_L \Rightarrow i_L$

Switching Losses

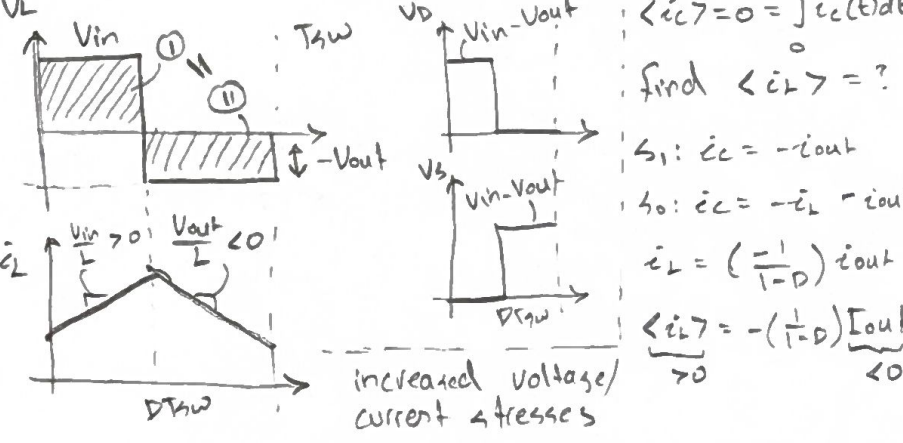
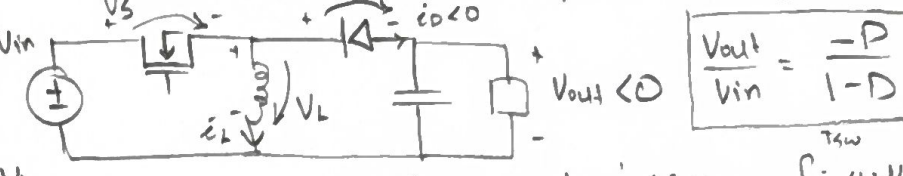


Cuk Converter: only use 2 pwr semi's, but penalty is negative polarity output voltage



$\langle V_L \rangle = 0 \Leftrightarrow \int_0^{T_{sw}} V_L(t) dt = 0$ $\frac{V_{out}}{V_{in}} = \frac{-D}{1-D}$

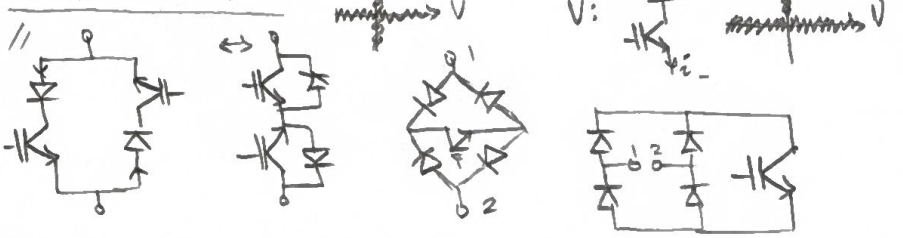
Inverting Buck-Boost: further reduce w magnet. comp. but each higher rating



Diode:



4-Quadrant Sw:



Conduc Losses:

$P_{cond} = I_{rms}^2 \cdot R_{on}$

$I_{rms} = \sqrt{\frac{1}{T_{sw}} \int_0^{T_{sw}} i^2(t) dt}$
 higher inductor ripple means higher conduc losses

Basics: $i_C = C \frac{dv_C}{dt}$, $V_L = L \frac{di_L}{dt}$

$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
 $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
 $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$
 $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$
 $\sin 2\theta = 2 \sin \theta \cos \theta$
 $2 \cos^2 \theta - 1 = \cos 2\theta = 1 - 2 \sin^2 \theta$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)] \quad \sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega t + \phi_n)$$

$$i(t) = I_0 + \sum_{m=1}^{\infty} I_m \cos(m\omega t + \theta_m)$$

$$V_n = \sqrt{a_n^2 + b_n^2} \quad a_n = \frac{2}{T} \int_0^T v(t) \cos(n\omega t) dt$$

$$b_n = \frac{2}{T} \int_0^T v(t) \sin(n\omega t) dt$$

instantaneous power: $p(t) = i(t)v(t)$

Average Power: $P_{av} = \frac{W_{total}}{T} = \frac{1}{T} \int_0^T i(t)v(t) dt$

- ① DC > DC: $V_0 I_0$
 - ② DC < sinusoid: 0
 - ③ sinusoid < sinusoid: 0 if $m \neq n$
- if $n=m$: $\frac{V_n I_n \cos(\phi_n - \theta_m)}{2}$

RMS in t: $I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$

RMS in freq: $I_{rms} = \sqrt{I_0^2 + \sum_{m=1}^{\infty} \frac{I_m^2}{2}}$

$P_{loss} = I_{rms}^2 R = \frac{V_{rms}^2}{R}$

$S = V_{rms} I_{rms}$

$P_{avg} = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$

PF angle θ

$P[\omega]$ lead pf (C) $\downarrow$

real pur [W] $\uparrow$ apparent pur [VA] $\uparrow$

$PF = \frac{P}{S}$

$PF = \frac{P_{avg}}{V_{rms} I_{rms}}$

THD: $\sqrt{\sum_{n=2}^{\infty} I_n^2} / I_1$

distortion factor $= \frac{1}{\sqrt{1+THD^2}}$

true PF = (distortion factor) (displacement factor)

$= \left(\frac{1}{\sqrt{1+THD^2}} \right) (\cos(\phi_1 - \theta_1))$

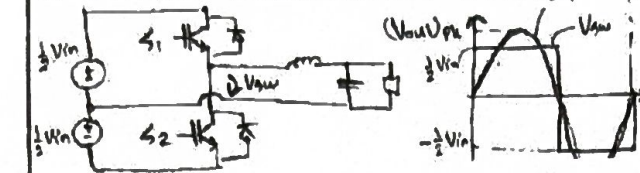
$V(t) = V_{rms} \sqrt{2} \cos(\omega t + \phi) \Rightarrow \bar{V} = V_{rms} \angle \phi$

$V_{rms} = \sqrt{V_0^2 + \sum_{n=1}^{\infty} \frac{V_n^2}{2}}$

$V_{rms} = \sqrt{\sum_{n=1}^{\infty} (V_{rms})_n^2}$

$R_{eq} = R_1 + R_2, X_{L2} = X_{L1} + X_{L2}$

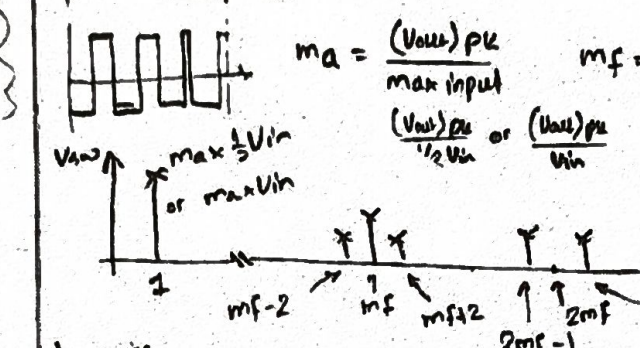
Single Phase Half-Bridge Inverter



$V_{out} = (V_{out})_{pk} \sin(2\pi f_{out} \cdot t) = \frac{4}{\pi} \cdot \frac{1}{2} V_{in} \sin(2\pi f_{out} \cdot t)$

S_1, S_2 can be off simultaneously but not both on.

SPWM: triangular carrier, rectangular V_{sw} , desired sinusoid $m(t)$

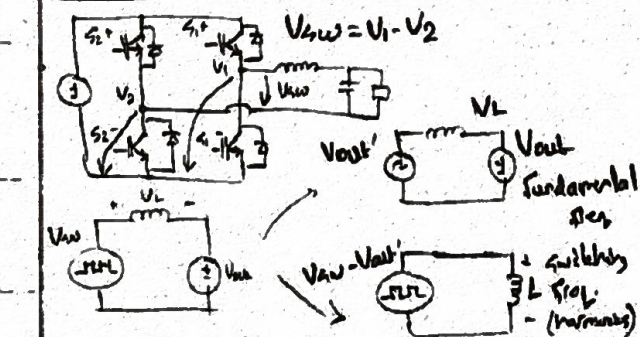


harmonics order: $j m_f \pm k$

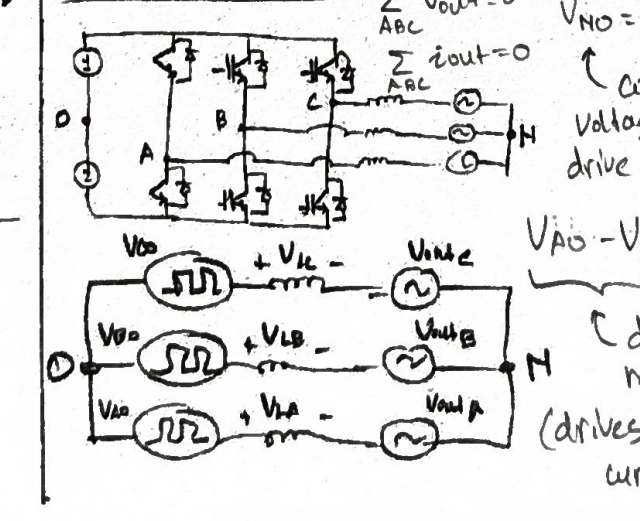
odd $j \rightarrow$ even $k \leftarrow$ preferred and happens when m_f is odd

even $j \rightarrow$ odd k

Single Phase Full Bridge Inverter



Three Phase Inverter



$F = N i [A \cdot \text{turns}] \quad B[T] = \mu H = \mu_0 \mu_r H \left[\frac{A}{m} \right]$

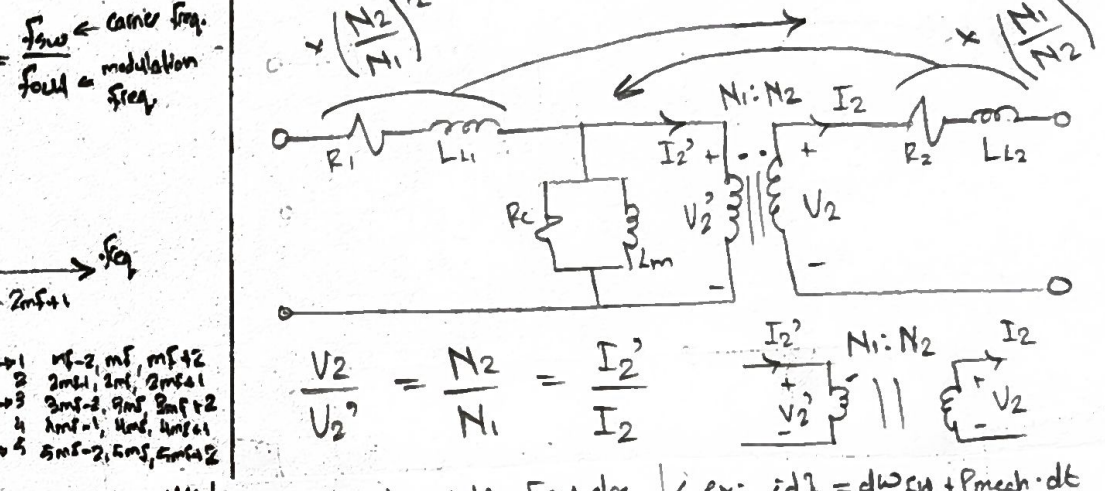
$F = \frac{\phi}{[Wb]} (R_c + R_g) = N i \quad R_c = \frac{l_c}{\mu_0 \mu_r A_c}$

$R_g = \frac{g}{\mu_0 A_g}$

$\mu_0 = 4\pi \times 10^{-7}$

$L = \frac{\lambda}{i} = \frac{N \phi}{i} = \frac{N^2}{R_c + R_g}$, if $\mu_r \rightarrow \infty, L = \frac{N^2}{R_g}$

$\omega = \frac{1}{2} L i^2 = \frac{1}{2} B H$ (Volume) of induct.



$dW_{fld}(\lambda, \epsilon) = \epsilon d\lambda - F_{fld} d\epsilon$

$\epsilon = \frac{\partial W_{fld}}{\partial \lambda} |_{\epsilon}, F_{fld} = -\frac{\partial W_{fld}}{\partial \epsilon} |_{\lambda}$

$W_{fld}(\lambda, \epsilon) = \frac{1}{2} \lambda^2 \epsilon, F_{fld} = \frac{\epsilon^2}{2} \frac{d}{d\lambda} \lambda(\epsilon)$

$dW'_{fld} = \lambda_1 d\epsilon_1 + \lambda_2 d\epsilon_2 + T_{fld} d\theta, T_{fld} = \frac{\partial W'_{fld}}{\partial \theta} |_{\epsilon_1, \epsilon_2}$

$W'_{fld}(\epsilon_1, \epsilon_2, \theta) = \frac{1}{2} L_{11}(\theta) \epsilon_1^2 + \frac{1}{2} L_{22}(\theta) \epsilon_2^2 + M(\theta) \epsilon_1 \epsilon_2$

$T_{fld} = \frac{\epsilon_1^2}{2} \frac{d}{d\theta} L_{11}(\theta) + \frac{\epsilon_2^2}{2} \frac{d}{d\theta} L_{22}(\theta) + \epsilon_1 \epsilon_2 \frac{d}{d\theta} M(\theta)$

OC: parallel branch $V_{oc}, I_{sc}, P_{oc}, Z = \frac{V_{oc}}{I_{sc}} \angle \theta$

$P_{oc} = V_{oc} I_{sc} \cos \theta = \frac{1}{R_c} - j \frac{1}{X_m}$

$R_c = \frac{V_{oc}^2}{P_{oc}}, X_{oc} = \sqrt{S_{oc}^2 - P_{oc}^2}$

$S_{oc} = V_{oc} I_{sc}, X_m = \frac{V_{oc}^2}{Q_{oc}}$

SC: series impedance $V_{sc}, I_{sc}, P_{sc}, Z = \frac{V_{sc}}{I_{sc}} \angle \theta$

$P_{sc} = V_{oc} I_{sc} \cos \theta = R_{eq} + j X_{eq}$

$|Z_{eq}| = \frac{V_{sc}}{I_{sc}}, X_{eq} = \sqrt{|Z_{eq}|^2 - R_{eq}^2}$

$R_{eq} = P_{sc} / I_{sc}^2$

machine scaling laws: $P = M \omega = M 2\pi n$, M torque, ω angular speed, n rot. speed

Lorentz Force: $F = I(\vec{L} \times \vec{B})$, $W_m = \frac{2 W_e}{\text{poles}}$

$i_a = \hat{I} \cos(\omega t)$
 $i_b = \hat{I} \cos(\omega t - 120^\circ)$
 $i_c = \hat{I} \cos(\omega t + 120^\circ)$

if $i_a = \hat{I} \cos(\omega t) \rightarrow \phi$

for 3-wire signs of i_b, i_c produces 120° pos for dia. current, 120° rot for ϕ